
Specialist Physical AI Need Not Be Opaque A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a rational
4 vision-language-action model family with a 19.2M convolutional policy and a
5 20.1M ViT policy, both built only from bilinear, linear, and rational learned opera-
6 tions. Their weights can therefore be reorganized into explicit interaction tensors
7 rather than treated only as an opaque checkpoint. A runtime operator audit of
8 the 20.1M ViT checkpoint and local numerical reconstructions verify its operator-
9 level form. Three historical aggregate records for convolutional χ -VLA report
10 93.7% $\pm$ 0.8% across LIBERO-Object under the recorded three-seed protocol. The
11 three released convolutional checkpoint files named in those historical aggregate
12 records also pass an independent 4,608-case joint-attention certificate with worst
13 relative error 6.36×10^{-16} . In a separate ViT instantiation, we reconstruct every
14 attention module in three checkpoints with worst relative error 2.56×10^{-7} . In a
15 seed-0 ViT checkpoint, exact weight-derived subspaces beat equal-rank random
16 controls in all 36 primary block-head tests. The result is a capable and structurally
17 inspectable specialist family with exact access to trained attention coefficients. It
18 does not yet provide an exact end-to-end decomposition of the full policy.

19 1 Introduction

20 Modern robot policies can achieve high success while giving little indication of why an action was
21 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
22 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
23 activations, yet a correlation is not automatically a mechanism or a safe editing target.

24 This is not a zero-shot policy. It is trained from scratch on one specialist suite. Its role in the
25 workshop’s question is to establish an auditable specialist control against which future zero-shot and
26 transfer claims can be evaluated.

27 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
28 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
29 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
30 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
31 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
32 tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one
33 compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For

example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We report three historical aggregate records for a fully rational 19.2M convolutional policy under a protocol aligned with published LIBERO-Object results. The records give $93.7\% \pm 0.8\%$.
3. We mechanically audit the 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

76 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 77 root. Let $m(x) = d^{-1} \sum_{j=1}^d x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We
 78 deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \quad \text{RNorm}(x) = \frac{x}{\sqrt{s_0}} r(v). \quad (3)$$

79 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 80 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 81 rescaling without leaving the rational tensor representation. Training and deployment use r itself. The
 82 exactness claims below therefore concern faithful evaluation of the deployed rational computation,
 83 not uniform agreement with RMSNorm. Appendix A.3 quantifies that distinction.

84 2.2 Architecture and training

85 Both encoder variants follow the high-level VLA sequence of visual encoding, language and state
 86 embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an
 87 eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share
 88 a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query
 89 state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision
 90 blocks and has 20,137,352 learned parameters. The 19.2M convolutional variant instead uses three
 91 stride-2 bilinear convolution stages that produce an 8×8 token grid. The historical capability records
 92 use the convolutional variant. The runtime operator census and reduced-Gram screen use the ViT. The
 93 released convolutional files additionally receive the joint-attention and rational-normalizer certificates
 94 reported in Appendix A.3.

95 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 96 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 97 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 98 that verify every requested state was loaded.

99 2.3 Mechanical certificate on the 20.1M ViT checkpoint

100 Architecture definitions can differ from deployed code. We therefore audit the real 20.1M ViT
 101 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

102 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 103 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 104 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
 105 summarized in Appendix Figure 4. It does not materialize one compact polynomial for the full
 106 eight-layer transformer.

107 3 Protocol-aligned closed-loop capability

108 3.1 Evaluation protocol

109 The historical aggregate records specify the following evaluation protocol for all ten LIBERO-Object
 110 tasks:

- 111 • 50 canonical trials per task
- 112 • a 280-step cap
- 113 • three independently trained convolutional seeds
- 114 • 500 rollouts per seed and 1,500 rollouts in total

115 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 116 rerun in our codebase, so differences may still include implementation and training choices.

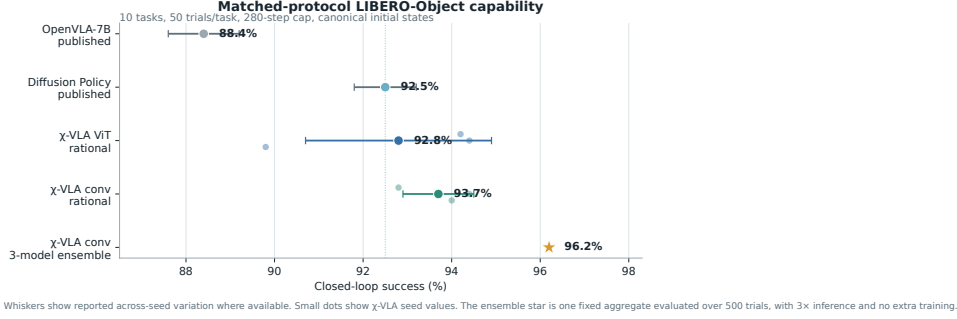


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are historical χ -VLA aggregate seed values. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	19.2M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 19.2\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The single-model χ -VLA row is the mean $\pm$ sample standard deviation from surviving historical aggregate records. Their original per-episode files are unavailable. The ensemble row comes from a separate paired evaluation. The two external rows are historical references rather than current state-of-the-art claims.

The three historical convolutional aggregate records report 94.0%, 94.4%, and 92.8%. Their narrow spread is descriptive bounded evidence across independently trained checkpoints.

Separately, the paired ensemble evaluation re-evaluates the three members at 93.6%, 95.0%, and 93.2%, for a 93.9% mean. Elementwise prediction averaging reaches 96.2%. This is 2.3 points above that paired mean and 1.2 points above its strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 19.2M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$

140 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
 141 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
 142 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
 143 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

144 We separately certify the three released convolutional checkpoint files named in the historical
 145 capability aggregate records. Their vision stacks contain no attention, so we independently rebuild all
 146 eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically
 147 selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every
 148 output is finite and the worst relative ℓ_2 error is 6.36×10^{-16} , far below the frozen 10^{-6} gate. This
 149 same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does
 150 not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or
 151 action head as one end-to-end contraction.

152 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
 153 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
 154 rank important directions without constructing the full table. It matches brute-force materialization to
 155 2.13×10^{-14} at toy scale.

156 4.2 Trained policy result

157 The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking
 158 experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived
 159 calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random
 160 projector banks are sampled once, then fixed and reused for all examples. The weights choose the
 161 subspace, while cached activations are used only afterward to score how well it preserves the layer’s
 162 computation.

163 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
 164 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
 165 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
 166 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent
 167 model-level replicates.

168 This extends exact weight-only construction through individual attention layers. It is not an exact
 169 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
 170 rapid degree and representation growth.

171 5 Related work and limitations

172 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
 173 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
 174 weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also
 175 steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived
 176 construction in a policy designed as a rational tensor program. Tensor and polynomial networks
 177 provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard
 178 LIBERO success can coexist with weak language use [9–11].

179 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
 180 robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce
 181 one compact symbolic contraction of the full policy. The released convolutional files support the
 182 joint-attention equation certificate and are named in the historical capability aggregate records. The
 183 reduced-Gram subspace screen uses separate ViT checkpoints. Evaluation uses fixed task instructions
 184 and does not establish robust instruction grounding. Within that scope, rational restrictions support
 185 strong LIBERO-Object control and exact trained-weight certificates. We do not claim a direct
 186 coefficient-edit interface.

187 **Reproducibility.** The anonymous artifact releases raw structural-certificate JSONs, immutable
 188 SHA-256 identities, and a standard-library one-command verifier. Historical Modal cache-backed
 189 analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata

190 and maps dataset identifiers to official identifiers before defining task splits. A source-level audit
191 compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned
192 LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched
193 exactly for every frame. Canonical-state checks fail loudly when requested states are unavailable.

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219 A Supporting results and decisions

220 A.1 Per-task ensemble behavior

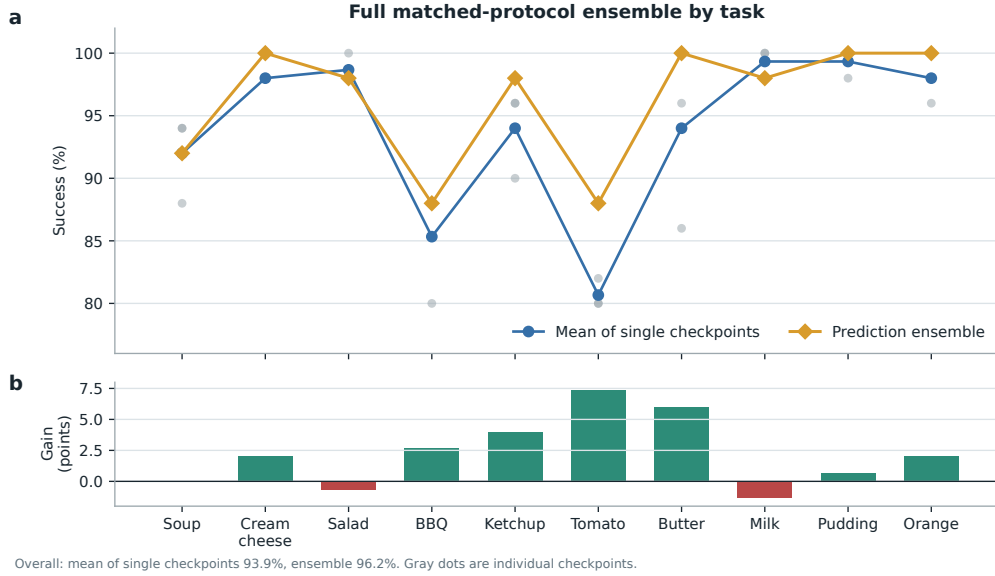


Figure 2: **Per-task effect of prediction averaging in the paired ensemble evaluation.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 19.2M-parameter policy.

221 A.2 Expanded attention screen

222 Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all
 223 eight blocks and twelve heads. At rank 128, 83/96 block-head pairs favor the exact weight-derived
 224 subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth
 225 controls. At ranks 8 and 32, only 54/96 and 60/96 pairs favor the exact subspace. The effect is
 226 therefore depth-dependent and strongest at the wider tested rank.

227 A.3 Numerical scope of rational conversion

228 The deployed degree- $[2/2]$ least-squares rational fit is exact as a rational computation even though it
 229 approximates RMS normalization. Its numerator has coefficients (5.32997, 7.53552, 0.327274) and
 230 the denominator has coefficients (1, 9.64827, 2.60664). Every denominator coefficient is positive,
 231 which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances,
 232 each invoked once per forward pass, on 4,096 deterministic inputs for each of three convolutional
 233 checkpoints. Across 544,542,720 activation rows, the minimum observed denominator is 1.0016 and
 234 no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that
 235 recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined
 236 as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3}
 237 gate.

238 The certificate also bounds what the operator approximates. Observed normalized mean-square ratios
 239 v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs,
 240 the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS
 241 scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The
 242 positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned
 243 floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS
 244 scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

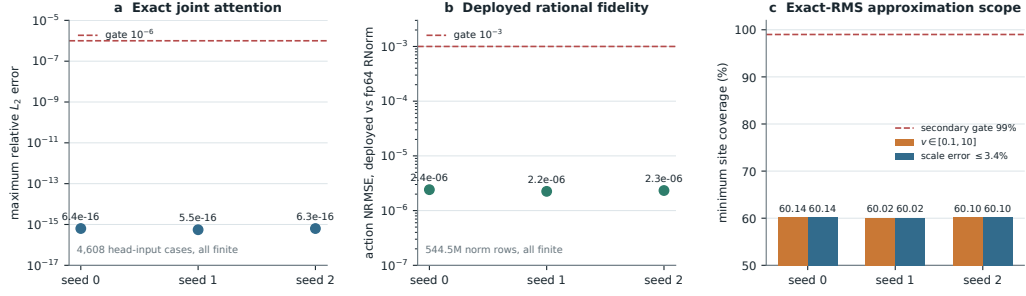


Figure 3: **Frozen structural certificates on the released convolutional files named in the historical capability aggregate records.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all 4,608 audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.

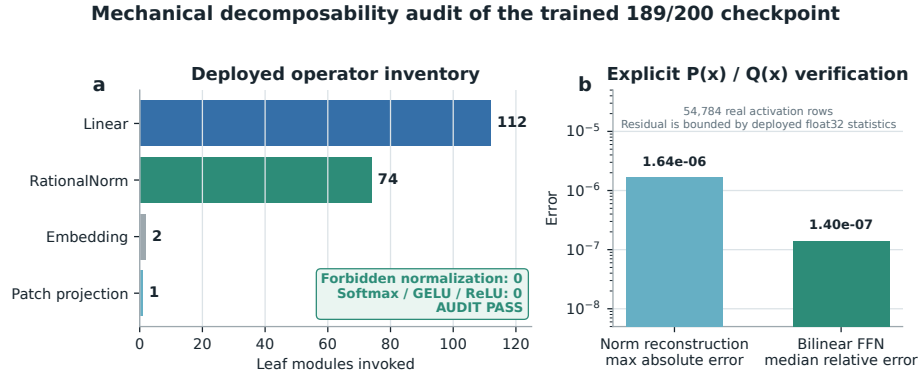


Figure 4: **Mechanical audit of the 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.